John Miller

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Experience

ExtraHop – cybersecurity threat detection and analysis

Senior Software Engineer AI/ML — AUG 2026 - Present

Senior Data Scientist — MAY 2024 - AUG 2026

Data Scientist III — SEP 2021 - MAY 2024

- Researching and architecting evaluation frameworks for graph-based agentic workflows supporting alert triage
- Built and productionalized an AWS pipeline that diffs daily TLD zone files to track domain age for reputation scoring
- Engineered a backtesting framework that replays incident-correlation templates against historical customer events
- Led development of a completely novel entropy-based ransomware detector, coordinating across multiple teams
- Constructed an AWS Glue pipeline capable of processing billions of records for domain and IP reputation scoring
- Wrote and reviewed customer-facing production code for ML-based real-time threat detection in network data
- Owned end-to-end development and deployment of an internal data analysis package for the data science team

Billups – geospatial analysis for marketing and advertising

Data Scientist — FEB 2020 - JUL 2021

- Overhauled company's campaign measurement pipeline, reducing necessary manual work from days to hours
- Implemented an automated PySpark workflow, decreasing computation time by 20% and minimizing human error
- Completed advertising campaign attribution and lift studies for dozens of clients across the United States

ZoomInfo – data enrichment and classification for business sales intelligence

Data Scientist — OCT 2018 - FEB 2020

- Collaborated with data engineers, product managers, and executive leadership to develop and implement machine learning and deep learning models for automated data cleaning and enrichment
- Trained a regression model to predict company employee counts, improving private company estimates by 15%

Garvin Image Lab – automated diagnostic tools for diseases of the retina

Masters Fellow — AUG 2016 - JUN 2018

- Developed novel combinations of shape analysis and image processing techniques for disease classification
- Trained random forest classifiers to determine the cause of optic disc swelling with 86% accuracy

Human Spinal Cord Research Lab – neuroscience research and experiment design

Research Assistant — SEP 2014 - AUG 2016

- Designed, conducted, and analyzed experiments investigating the therapeutic effects of spinal cord stimulation
- Wrote and maintained Matlab codebase for the research team's analysis and visualization workflows

Projects

LyricsGenius – python wrapper for downloading song lyrics and metadata from genius.com

Creator & Maintainer

- Actively maintaining package on GitHub for over 30k users, coordinating open-source collaboration
- *Trucks and Beer* music lyrics analysis project awarded "best data-driven stories of 2018" by *The Pudding*

Muscle MIDI Music – arduino / iPhone device for translating muscle activity into electronic music

Sole Developer

- Designed and built an electrical circuit and wrote software that generated and controlled music via muscle signals
- Winner out of 200+ entrants in the "Sensors Contest 2017" on Instructables.com

Skills

Code: Python, MLflow, LangChain, pandas, scikit-learn, SQL, PySpark, Matlab, NumPy, Jupyter, Flask, Matplotlib, Altair

Tools: AWS (SageMaker, Glue, DynamoDB, Athena, QuickSight, Lambda, S3, Fargate), Databricks, Git, GitHub, Docker

Topics: Machine learning, agentic systems, LLM evaluation, natural language processing, data visualization

Education

University of Iowa – Master of Science in Electrical and Computer Engineering — 2018

Goshen College – Bachelor of Arts in Physics, minor in English — 2014